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HCI PROBLEM FRAMING

TeachBack: Explain It to Another Student

Education Theme • Skill Development • HCI Project

Design opportunity: Use “teaching another student” as an interaction for learning. Instead of only consuming explanations or answering questions, students explain a concept to a virtual peer who asks follow-up questions, reveals misunderstandings, and helps the learner reflect on gaps in their understanding. Critically, the virtual peer does not simulate confusion at random: it forms a visible model of the concept from what the student has said, so that an incomplete explanation produces a visibly incomplete understanding that the student must repair.

1. THE PROBLEM

Students can encounter a gap between recognising information and being able to explain it clearly. A learner may feel that a concept is understood while still being unable to describe the idea, connect its parts, answer a “why?” question, or correct a misunderstanding.

This gap is difficult for students to detect on their own. Re-reading notes produces a sense of familiarity that is easily mistaken for understanding, and that impression is usually only corrected under assessment conditions, when it is too late to act on. Traditional study interfaces often emphasise reading, recall, and right-or-wrong answers, giving students limited opportunity to test their understanding by explaining a concept in their own words.

2. TARGET USERS

Primary users are university students who are learning conceptual or problem-solving subjects and want a way to check whether they genuinely understand what they have studied. The initial prototype will focus on individual study sessions, while the interaction is designed to simulate the experience of explaining a concept to a peer.

To keep the scope tractable, the prototype will target a small set of introductory computer science concepts with well-documented and predictable misconceptions, such as recursion, pointers and references, and binary trees. Concepts of this kind suit the design because the virtual student requires plausible incorrect beliefs to hold, and these are already documented in the computing education literature.

3. USER NEED

Students need a low-pressure way to practise explaining concepts, receive useful follow-up questions, expose gaps in their understanding, and reflect on what they know without simply being given the answer.

This need will be grounded rather than assumed. We will conduct short sessions with five to six students in which each participant rates their confidence across several topics from a current course and is then asked to explain a high-confidence topic aloud. The difference between stated confidence and the explanation produced serves as direct evidence of the gap described above, and the points at which explanations break down will inform the virtual student's questions.

4. PROPOSED DESIGN DIRECTION

TeachBack is a learning interface in which the student teaches a concept to a virtual student. The virtual student behaves like a learner rather than a conventional answer-giving chatbot. It can ask clarification questions, request examples, express confusion, and make plausible mistakes that the user must correct.

The central mechanism is inheritance: the virtual student adopts the understanding it is given, including its errors and omissions. If a concept is explained imprecisely, the virtual student comes to believe something imprecise, and the consequences become visible to the user. Alongside the dialogue, the interface maintains a representation of the virtual student's current understanding, so that the learner can see what has been successfully conveyed and what remains unclear or incorrect. After the interaction, the system provides a reflection summary highlighting concepts that were explained clearly and areas that may need more attention.

5. HCI FOCUS & DESIGN QUESTIONS

The project will investigate how a learner's understanding can be made visible and contestable through the act of teaching, and how conversational interaction can make “learning by teaching” usable, engaging, and supportive.

Key questions include: How should a learner's understanding be represented so that gaps are visible at a glance rather than buried in dialogue? How should the virtual student ask questions without frustrating the learner? How can the interface encourage explanation rather than answer-copying? How should feedback be presented without pretending that an automated judgement is always correct? A further question concerns tone: the interaction collapses into assessment if the virtual student reads as an examiner, so the design must sustain the impression of a peer.

6. RELATED WORK & POSITIONING

Learning by teaching is an established finding in education research, commonly described as the protégé effect, in which preparing to teach produces deeper engagement with material than preparing to be tested. The approach has been explored in teachable agent systems, most notably Betty's Brain, in which students teach a virtual agent by constructing a concept map and then observe it reason with what it has been taught.

TeachBack is positioned as an interaction design contribution rather than a replication. The focus is on natural explanation as the input modality rather than structured map construction, and on the design of the representation, pacing, and tone through which a learner's gaps are surfaced back to them.

7. SCOPE & SUCCESS

The prototype will demonstrate a focused teaching session around selected academic concepts rather than attempting to support every subject or student need. Success will be judged by whether the interaction clearly communicates the concept, encourages the student to explain in their own words, exposes a meaningful knowledge gap, and gives understandable next-step feedback.

Evaluation will take the form of think-aloud sessions with students from the target group, examining whether participants recognise a gap they had not previously identified, whether the virtual student's questions are experienced as helpful rather than obstructive, and whether the reflection summary is considered actionable.

8. ETHICS, SAFETY & INCLUSIVITY

Accessibility, privacy, bias, and the risk of misleading feedback will be considered throughout the design. The most significant risk is specific to this concept: a virtual student that models a concept incorrectly may reinforce a misconception rather than correct it. The design will therefore present the system's judgements as provisional and open to challenge, and will scope content to concepts where correctness can be bounded.

Explaining aloud can also be exposing, particularly for students who are less confident or who are working in a language other than their first. The interaction is designed to be private by default and unhurried, and must not read as evaluative. Records of what a student failed to explain are sensitive and should remain under the student's control.

Problem-framing hypothesis: If students are given an interactive peer-like audience that asks them to clarify and defend their explanations, and can see their explanation reflected back in what that audience comes to understand, they may become better able to identify and address gaps in their own understanding.